

Multiple Choice (10 marks)

1. Suppose your model is overfitting. Which of the following is NOT a valid way to try and reduce the overfitting?
 - (a) Increase the amount of training data.
 - (b) Improve the optimisation algorithm being used for error minimisation.
 - (c) Decrease the model complexity.
 - (d) Reduce the noise in the training data.
2. The disadvantage of Grid search is
 - (a) It can not be applied to non-differentiable functions.
 - (b) It can not be applied to non-continuous functions.
 - (c) It is hard to implement.
 - (d) It runs reasonably slow for multiple linear regression.
3. Which of the following sentence is FALSE regarding regression?
 - (a) It relates inputs to outputs.
 - (b) It is used for prediction.
 - (c) It may be used for interpretation.
 - (d) It discovers causal relationships
4. Statement 1| As of 2020, some models attain greater than 98% accuracy on CIFAR-10.
Statement 2| The original ResNets were not optimized with the Adam optimizer.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
5. Which among the following prevents overfitting when we perform bagging?
 - (a) The use of sampling with replacement as the sampling technique
 - (b) The use of weak classifiers
 - (c) The use of classification algorithms which are not prone to overfitting
 - (d) The practice of validation performed on every classifier trained

True / False (10 marks)

Answer True or False

6. True or False: Pruning a Decision Tree mainly aims to reduce the training set error by increasing the model's complexity.
7. True or False: The convergence of a neural network is influenced by the choice of learning rate during training.
8. True or False: A Bayesian network with the structure $H \rightarrow U \leftarrow P \leftarrow W$ requires only 4 independent parameters for its complete specification.
9. True or False: Frequentists support the use of prior distributions on parameters in probabilistic models, similar to Bayesians.
10. True or False: In Naive Bayes, attributes are statistically dependent of one another given the class value.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to calculate the height of the given binary tree.
12. Write a function to count the elements in a list until an element is a tuple.

End of Paper